

- This exam contains 10 pages (including this cover page) and 13 problems. Check to see if any pages are missing.
- You have until Monday, June 6 at 9:10 AM to complete this exam. You may use your notes and your book. You may not use the internet (aside from accessing MyMathLab or the WyoCourses page) for help on this exam. You may not collaborate with anyone else on this exam.

The following rules apply:

- If a question is followed by work, then work of some form is required. An incorrect answer supported by substantially correct calculations and explanations will receive partial credit. A correct answer that has no supporting work when work is required will not receive full credit.
- Organize your work, in a reasonably neat and coherent way, in the space provided. Work scattered all over the page without a clear ordering will receive very little to no credit.
- Although there are 162 points total, this is a 150 point exam. In other words, I will essentially not count the equivalent of one incorrect 15 point question against your score. Grades above 100% will not be given.
- If a question has multiple parts, the points for that question are divided evenly among the parts.
- Do not write in the $\checkmark$ + - box, because this is for grading. The grading scale (except True/False) is: $\checkmark$ = 100%, + = 50%, - = 0%

Page:	3	4	5	6	7	8	Total
Points:	20	30	40	15	15	42	162
Score:							

1. (10 points) A normal distribution has a mean of μ and a standard deviation of 12. If the 84th percentile is $P_{84} = 98$, then find the mean μ .

2. (10 points) In a normal distribution with a mean of 18, the number 25 has a standardized value of 1.6. What is the standard deviation of this normal distribution?

3. (15 points) The actual weight of the contents of an 11-ounce bag of potato chips has an approximate normal distribution with a mean of 11.6-ounces and a standard deviation of 0.2-ounces. What is the probability of randomly picking a bag with an actual weight greater than 11.8-ounces? Give your answer as a percentage.

4. (15 points) The battery life of an iPod has an approximate normal distribution with a mean of 6 hours and a standard deviation of 0.5 hours. What is the probability that a randomly chosen iPod will have a battery life between 5 hours and 6.5 hours? Give your answer as a percentage.

5. (15 points) An honest coin is tossed a total of $n = 1600$ times and the number of heads obtained is recorded. What is the probability that the coin lands heads greater than 840 times? Give your answer as a percentage.

6. (25 points) After finishing college, you start working as a quality control agent at a local manufacturing plant. The machine that you are in charge of produces steel bolts. Your job is to fine tune the machine so that it produces a specific size bolt, but a small margin of error is allowable. Suppose that you believe that the machine has been calibrated so that the actual size of a bolt produced has an approximate normal distribution with a mean of 25 cm and a standard deviation of 0.001 cm. After producing 1000 bolts, you randomly choose one and find that it measures 25.008 cm. Based off of this measurement, do you think that the machine has been properly calibrated? Explain.

7. (5 points) A normal distribution has a mean of μ and a standard deviation of σ . Suppose that you know that the 2.5th percentile, $P_{2.5} = 8$, and the 84th percentile, $P_{84} = 29$. Find the mean of this normal distribution.

- (a) 7
- (b) 13
- (c) 18
- (d) 22
- (e) None of the above.

8. (5 points) In a normal distribution with a standard deviation of 8, the number 25 has a standardized value of 2.5. What is the 84th percentile, P_{84} of this normal distribution?

- (a) 45
- (b) 53
- (c) 62
- (d) 84
- (e) None of the above.

9. (5 points) The height of an 18-year old student at your university has an approximate normal distribution with a mean of 68 inches and a standard deviation of 2.5 inches. What is the third quartile of height in this distribution? Round your answer to two decimal places.

- (a) 63 inches
- (b) 65.5 inches
- (c) 69.69 inches
- (d) 72.42 inches
- (e) None of the above.

10. (5 points) Which of the following scenarios would it not be appropriate to model with a normal distribution?

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- (a) Height of an 18-year old male student at your university.
- (b) Actual weight of the contents of a 12-ounce can of dog food.
- (c) Number of fingers on the left hand of an 18-year female student at your university.
- (d) Verbal SAT score of a recent high school graduate.
- (e) None of the above.

11. (10 points) An honest six-sided die is tossed a total of $n = 324$ times and the number of times that the die lands on an even number (2, 4, or 6) is recorded. What is the probability that between 162 and 171 of the tosses result in an even number? Give your answer as a percentage.

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12. (42 points) Short answer questions. Provide a brief sketch of your work to support your answer if computation is necessary.

(a) ☐ ☒ ☐ Explain why a distribution with $\mu = 240$, $Q_1 = 225$, and $Q_3 = 260$ cannot be a normal distribution.

(b) ☐ ☒ ☐ A normal distribution has mean $\mu = 100\text{kg}$ and standard deviation $\sigma = 10\text{kg}$. Find the z -value for $x = 123\text{kg}$.

(c) ☐ ☒ ☐ A normal distribution has mean $\mu = 100$ points and first quartile $Q_1 = 93.25$ points. Find the z -value for $x = 82$ points.

- (d) ☐ ☒ ☐ What is the z -value of Q_1 in any normal distribution?
- (e) ☐ ☒ ☐ A normal distribution has mean $\mu = 63.4$ and the 16th percentile of the distribution is $P_{16} = 53.8$. What is the standard deviation? Calculate P_{84} .
- (f) ☐ ☒ ☐ Suppose that a random sample of $n = 6084$ adults is to be chosen for a survey. Assume that the gender of each adult in the sample is equally likely to be male as it is female. Estimate the probability that:
- the number of females in the sample is between 3003 and 3081.
 - the number of females in the sample is less than 3003.
 - the percentage of females in the sample is below 50.64%.

13. (10 points (bonus)) Find a MST and a MaxST in the following network:

